

Agenda

- Classification Models
- Logistic Regression Algorithm
- Sklearn Logistic Regression
- Evaluating Classification Models Performance

1. Classification Models

Classification:

- Classification is a supervised learning concept which basically categorizes a set of data into classes.
- It is a predictive modelling problem where a class label is predicted for a given example of input data.

Classification problem try to learn categorical class.
such as “red” or “blue” or “yellow”

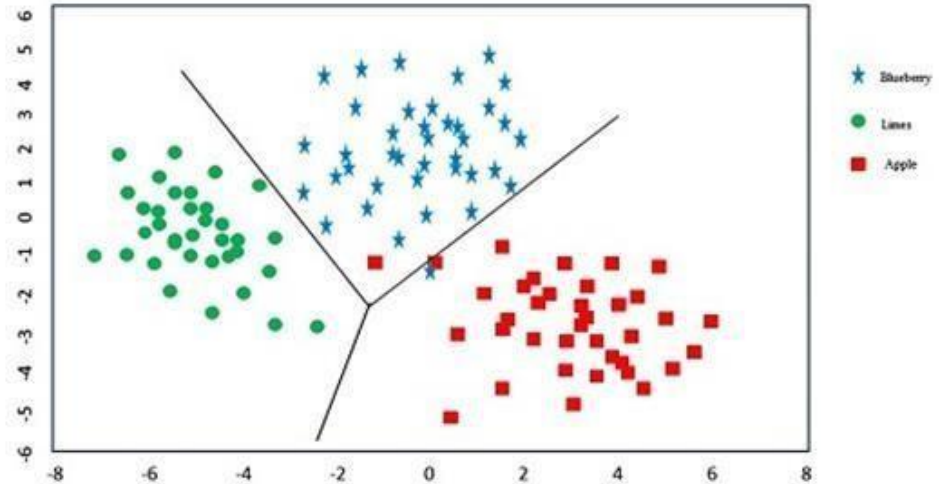
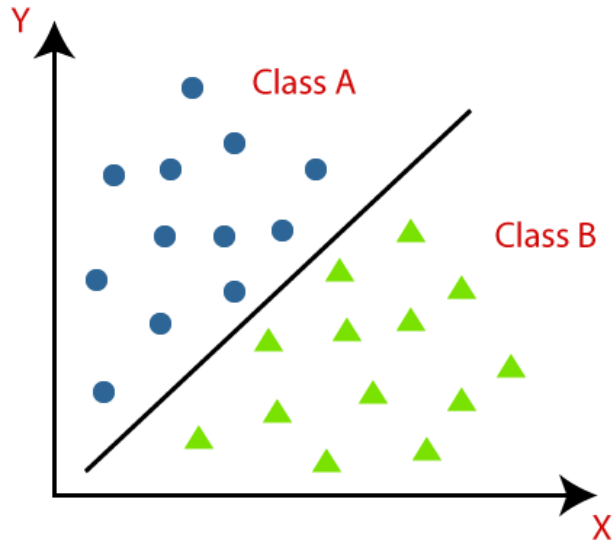
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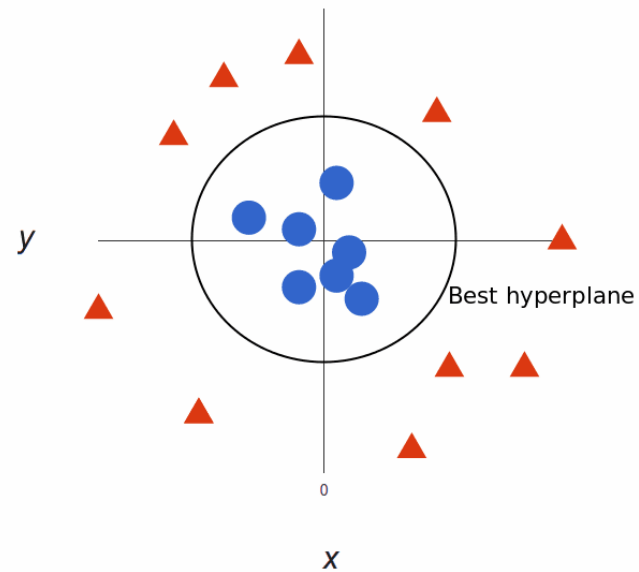
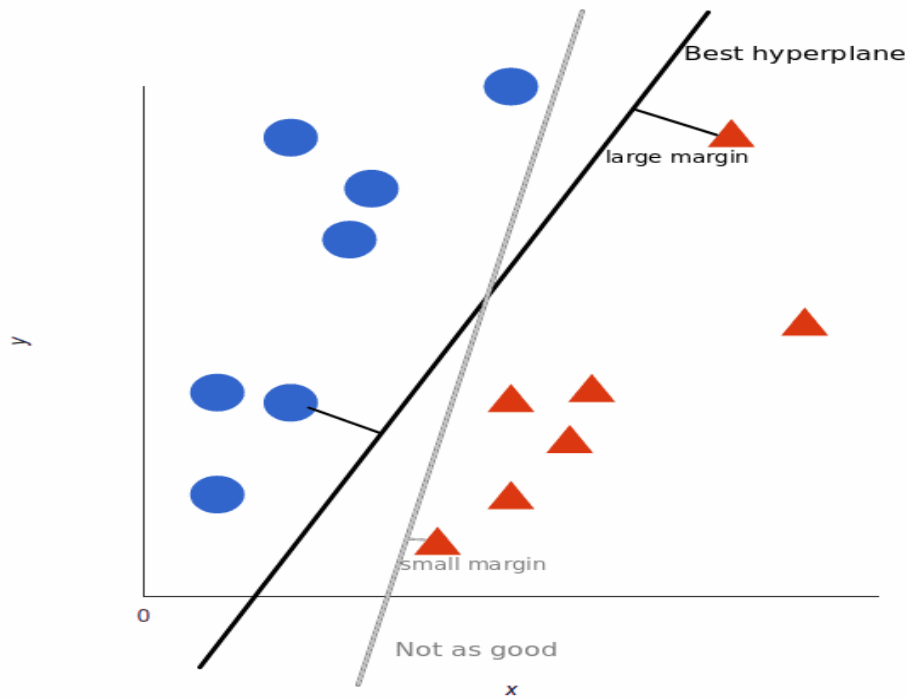
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Classification :

→ Classification algorithm is a Supervised Learning technique that is used to identify the class of new observations on the basis of training data.



Classification :



Classification use case :

Multi-Class Classification



Cat
Dog
Fox
Tiger
Lion



0 5
1 6
2 7
3 8
4 9



Person A
Person B
Person C



Java
C++
Python
C#
Go

Multi-Label Classification



Action
Crime
Thriller
Comedy
Drama



Cat
Dog
Tiger
Bird
Fish

Classification



Spam
Not Spam



Cancer
Not Cancer



Positive Sentiment
Negative Sentiment



Fraud
Not Fraud

Binary Classification

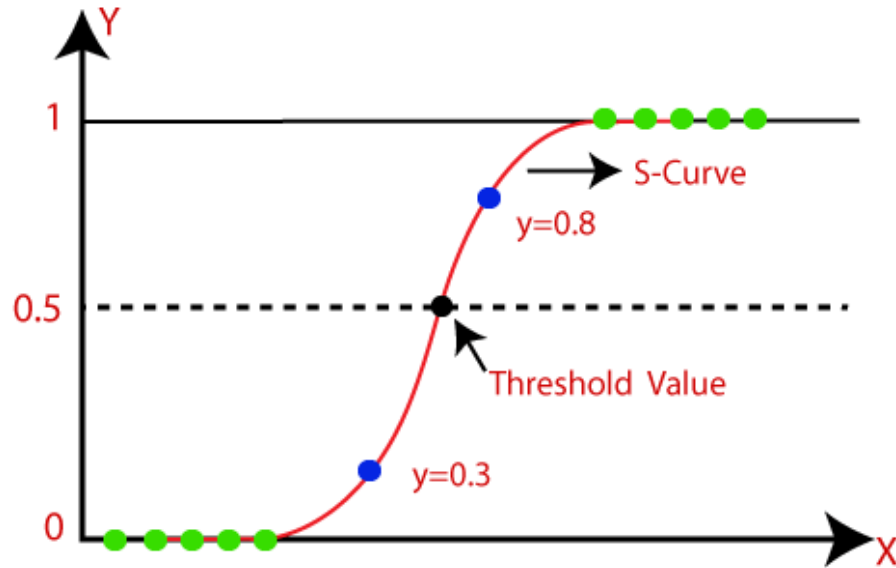
2. Logistic Regression Algorithm

Logistic Regression:

- **Logistic Regression** is a supervised learning classification algorithm used to predict the probability of a **Target Categorical Variable**.
- The outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, True or False, etc. but instead of giving the exact value as 0 and 1, it gives the **Probabilistic Values** which lie between 0 and 1.
- In Logistic Regression, instead of fitting a regression line, we fit an **"S"** shaped logistic function, which predicts two maximum values (0 or 1).
- The curve from the logistic function indicates the likelihood of something.

Logistic Function:

→ The **Logistic/Sigmoid** function is a mathematical function used to map the predicted values to probabilities.



Logistic Regression Equation:

- The below is the Hypothesis of Linear Regression.
- Logistic Regression just has a transformation based on it.
- For logistic regression, focusing on binary classification here, we have class 0 and class 1.

$$h_{\theta}(x) = \theta_0 x_0 + \theta_1 x_1 + \dots + \theta_j x_j = \theta^T x = \begin{bmatrix} \theta_0 & \theta_1 & \dots & \theta_j \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_j \end{bmatrix}$$

Where $x_0 = 1$

Logistic Regression Equation:

$$\hat{Y} = h_{\theta}(x) = \sigma(\theta^T \cdot x)$$

$\hat{Y}$: probability that an instance x is belongs to a class

h_{θ} : model parameters that determine boundary decision

x : features

σ : logistic (sigmoid — Logit) S-shaped function

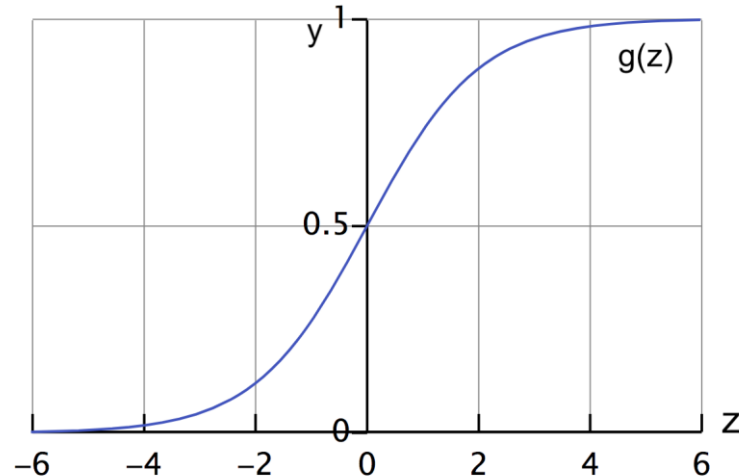
- Map any real-valued number to between 0, 1

Logistic Regression Equation:

- To compare with the target, we want to constrain predictions to some values between 0 and 1.
- That's why **Sigmoid Function** is applied on the raw model output and provides the ability to predict with probability.

Sigmoid Function : $g(z) = \frac{1}{1 + e^{(-z)}}$

Hypothesis : $h_{\theta}(x) = \frac{1}{1 + e^{(-\theta^T x)}}$



Logistic Regression Equation:

- What hypothesis function returns is the probability that $y = 1$, given x , parameterized by θ .
- Written as: $h(x) = P(y = 1 | x; \theta)$.
- Decision boundary can be described as:
Predict 1, if $\theta^T x \geq 0 \rightarrow h(x) \geq 0.5$ Predict
0, if $\theta^T x < 0 \rightarrow h(x) < 0.5$

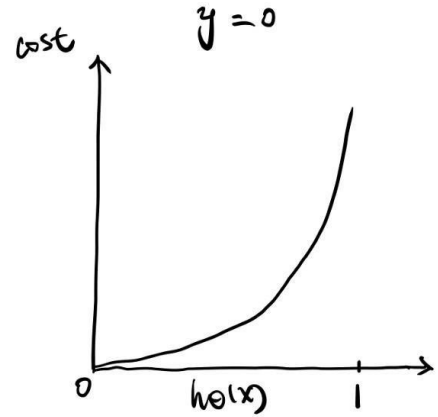
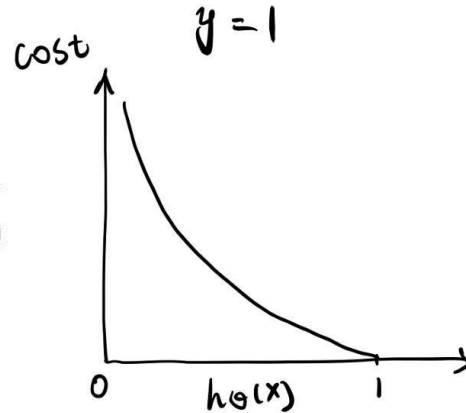
Cost Function:

- We want to assign more punishment when predicting 1 while the actual is 0, and when predicting 0 while the actual is 1.
- The Loss Function of Logistic Regression is doing this exactly which is called **Logistic Loss**.

Cost Function:

- If $y = 1$, the plot on left, when prediction = 1, the cost = 0, when prediction = 0, the learning algorithm is punished by a very large cost.
- Similarly, if $y = 0$, the plot on right, predicting 0 has no punishment but predicting 1 has a large value of cost.

$$Cost(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



Cost Function:

→ If $y = 1$, the plot on left, when prediction = 1, the cost = 0, when prediction = 0, the learning algorithm is punished by a very large cost

$$Cost(h_{\theta}(x), y) = -y \log(h_{\theta}(x)) - (1 - y) \log(1 - h_{\theta}(x))$$

$$\begin{aligned} J(\theta) &= \frac{1}{m} \sum_{i=1}^m Cost(h_{\theta}(x^{(i)}), y^{(i)}) \\ &= -\frac{1}{m} \left(\sum_{i=1}^m y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right) \end{aligned}$$

Optimization Function:

- We can start to fit by minimizing $J(\theta)$ as a function of θ to find optimal parameters.
- We can still apply **Gradient Descent** as the optimization algorithm.
- It takes partial derivative of J with respect to θ (the slope of J), and updates θ via each iteration with a selected learning rate α until the **Gradient Descent** has converged.

$$\theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

3. Sklearn Logistic Regression

Sklearn Logistic Regression:



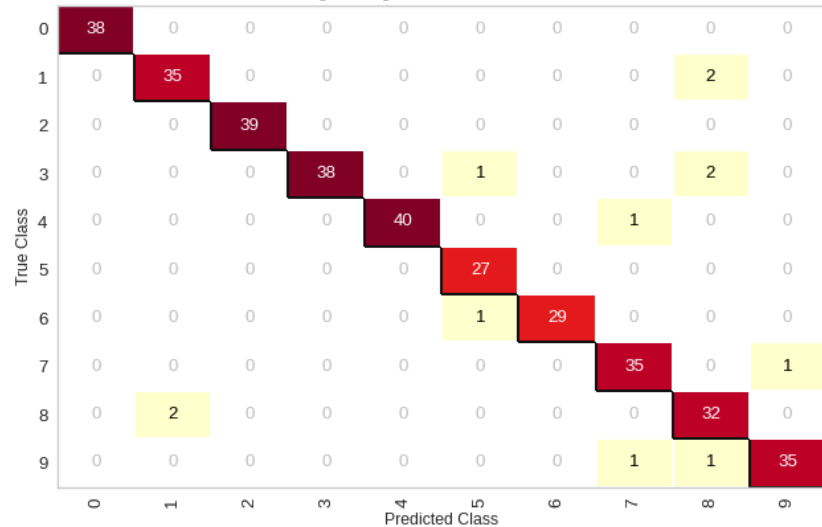
4. Evaluating Classification Models **Performance**

Evaluating Classification Models Performance

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1 Score

Evaluating Classification Models Performance

		Actual class	
		Positive	Negative
Predicted class	Positive	TP: True Positive	FP: False Positive (Type I Error)
	Negative	FN: False Negative (Type II Error)	TN: True Negative



Confusion Matrix:

There are four quadrants in the confusion matrix, which are symbolized as below.

True Positive (TP: f_{++}): The number of instances that were positive (+) and correctly classified as positive (+v).

True Negative (TN: f_{--}): The number of instances that were negative (-) and correctly classified as (-).

False Negative (FN: f_{+-}): The number of instances that were positive (+) and incorrectly classified as negative (-). It is also known as **Type 2 Error**.

False Positive (FP: f_{-+}): The number of instances that were negative (-) and incorrectly classified as (+). This also known as **Type 1 Error**.

		Actual class	
		Positive	Negative
Predicted class	Positive	TP: True Positive	FP: False Positive (Type I Error)
	Negative	FN: False Negative (Type II Error)	TN: True Negative

Accuracy:

This term tells us how many right classifications were made out of all the classifications. In other words, how many TPs and TNs were done out of TP + TN + FP + FNs. It tells the ratio of “True” s to the sum of “True” s and “False” s.

		Actual class	
		Positive	Negative
Predicted class	Positive	TP: True Positive	FP: False Positive (Type I Error)
	Negative	FN: False Negative (Type II Error)	TN: True Negative

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall (Sensitivity):

Out of all the actual real positive cases, how many were identified as positive.

		Actual class	
		Positive	Negative
Predicted class	Positive	TP: True Positive	FP: False Positive (Type I Error)
	Negative	FN: False Negative (Type II Error)	TN: True Negative

$$\begin{aligned}\text{Recall} &= \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \\ &= \frac{\text{True Positive}}{\text{Total Actual Positive}}\end{aligned}$$

Precision:

Out of all that were marked as positive, how many are actually truly positive.

		Actual class	
		Positive	Negative
Predicted class	Positive	TP: True Positive	FP: False Positive (Type I Error)
	Negative	FN: False Negative (Type II Error)	TN: True Negative

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

Precision:

F1 score is a weighted average of Precision and Recall, which means there is equal importance given to FP and FN. This is a very useful metric compared to **"Accuracy"**.

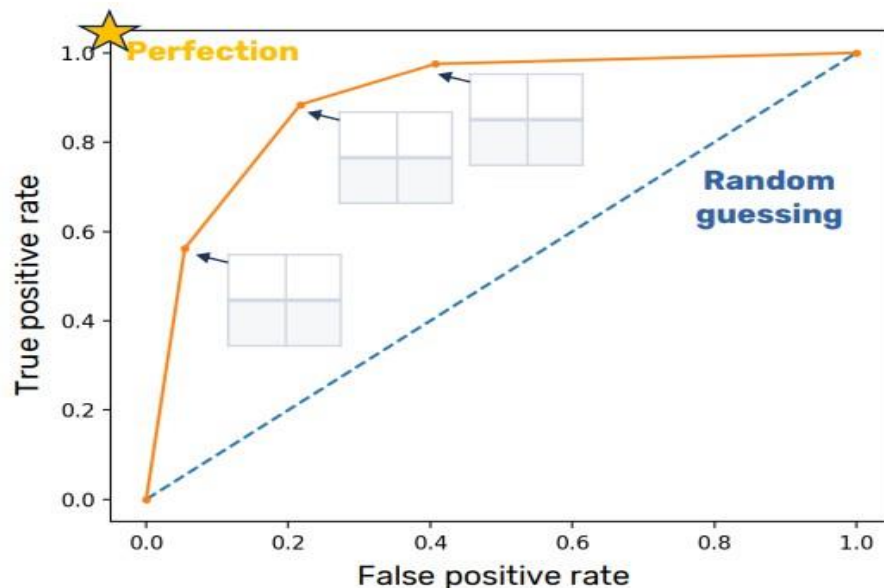
		Actual class	
		Positive	Negative
Predicted class	Positive	TP: True Positive	FP: False Positive (Type I Error)
	Negative	FN: False Negative (Type II Error)	TN: True Negative

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

$$\text{F1 Score} = \frac{\text{TP}}{\text{TP} + \frac{1}{2} (\text{FP} + \text{FN})}$$

Area Under Curve (AUC) and ROC Curve:

A ROC Curve is drawn by plotting TPR or True Positive Rate or Recall or Sensitivity (which we saw above) in the y-axis against FPR or False Positive Rate in the x-axis. $FPR = 1 - \text{Specificity}$ (which we saw above).



Confusion Matrix:

- Recall: Ability of a model to find all the relevant cases within a dataset.
- Precision: Ability of a model to identify only the relevant data points.
- F1 Score: harmonic mean of precision and recall.

Confusion Matrix:

		Actual class		
		Positive	Negative	
Predicted class	Positive	TP: True Positive	FP: False Positive (Type I Error)	Precision: $\frac{TP}{TP + FP}$
	Negative	FN: False Negative (Type II Error)	TN: True Negative	Negative Predictive Value: $\frac{TN}{TN + FN}$
		Recall or Sensitivity: $\frac{TP}{TP + FN}$	Specificity: $\frac{TN}{TN + FP}$	Accuracy: $\frac{TP + TN}{TP + TN + FP + FN}$

Case Study:

Case 1



COVID 19/ Healthy

Cost of FN > Cost of FP

Recall

$$\text{recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Case 2



Spam/Not Spam

Cost of FP > Cost of FN

Precision

$$\text{precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

Case 3



Good/Bad loan

Cost of FN > Cost of FP

Recall

$$\text{recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Any Questions?